Course Project

Data Analytics Foundation

Business Performance Dashboard

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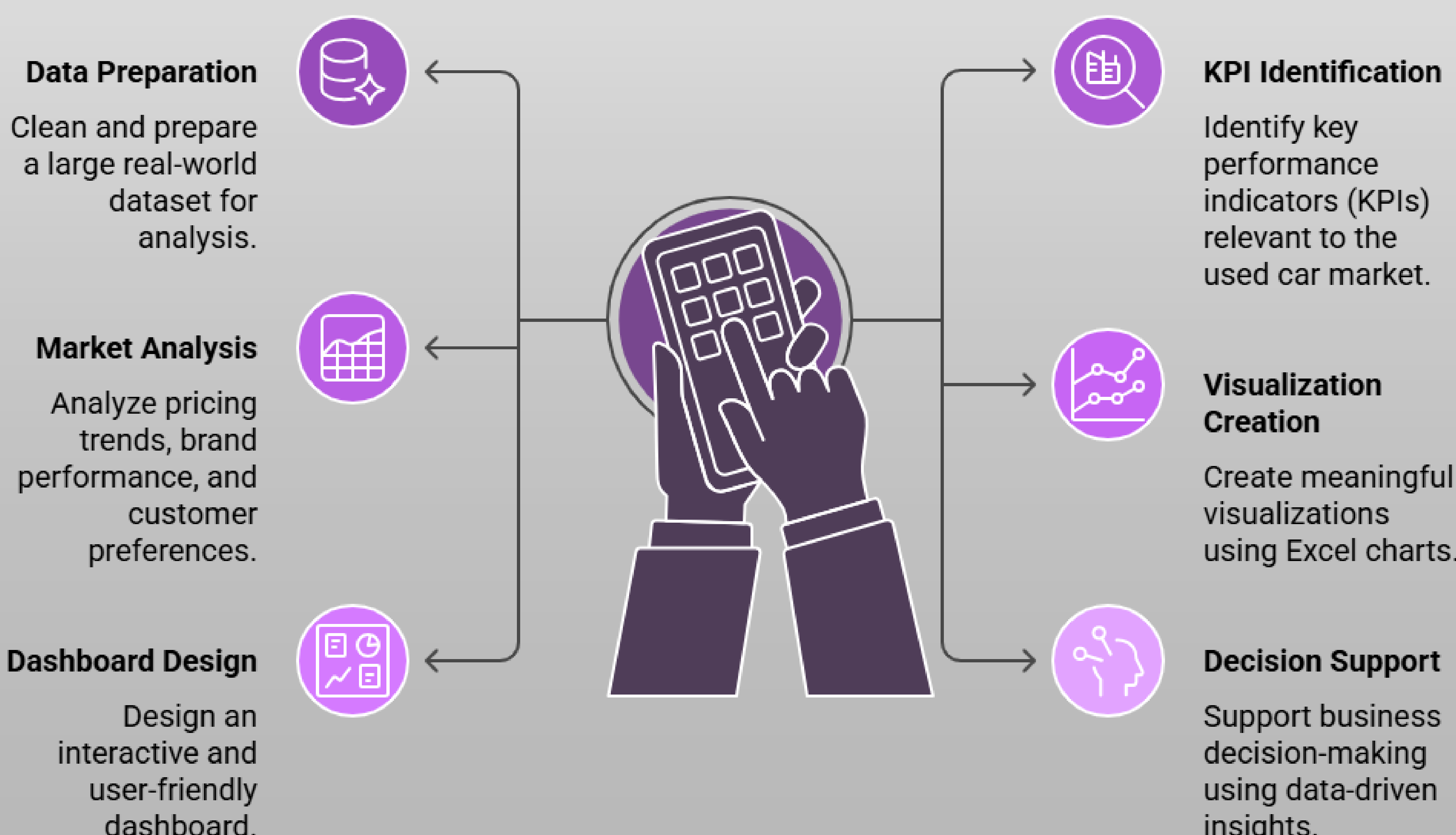
➢ Microsoft Excel Dashboard

1. Introduction –

- In today's data-driven world, organizations rely heavily on data analysis to make informed business decisions. The automobile resale market generates large volumes of data related to pricing, customer preferences, vehicle condition, and regional demand. Analyzing this data helps businesses understand market trends, optimize pricing strategies, and improve decision-making.

- This project focuses on analyzing a **used car sales dataset obtained from Kaggle** using **Microsoft Excel**. The aim is to clean, analyze, and visualize the data to extract meaningful business insights through KPIs, pivot tables, charts, and an interactive dashboard.

2. Objectives of the Project –



3. Dataset Description –

3.1 Dataset Source :

i. Source – Kaggle
ii. Type – Used Car Sales Dataset

3.2 Dataset Size :

i. Number of Rows – 60,000 (Approx)
ii. Number of Columns – 16

3.3 Dataset Attributes –

Column Name	**Description**
Car ID	Unique identifier for each car
Brand	Manufacturer of the car
Model Name	Car model
Model Variant	Variant of the model
Car Type	SUV, Sedan, Hatchback, etc.
Transmission	Manual / Automatic
Fuel Type	Petrol, Diesel, CNG, etc.
Year	Manufacturing year
Kilometers	Distance driven
Owner	Ownership type
State	Location of sale
Accidental	Accident history
Price	Selling price
Car Age	Calculated vehicle age
Price Range	Low / Mid / High
KM Category	Usage category

4. Data Cleaning and Preparation –

- Before analysis, the dataset was cleaned and transformed to ensure accuracy and consistency.

4.1 Data Cleaning Steps –

i. Removed duplicate records

ii. Removed rows with missing or zero price values

iii. Checked for blank values in critical columns such as Brand, Year, and Price

iv. Ensured correct data types (numeric, text, currency)

4.2 Feature Engineering (New Columns Created) –

4.2.1. Car Age – Car age was calculated to analyze depreciation –

✓ Car Age = Current Year – Manufacturing Year

4.2.2 Price Range – Cars were categorized into pricing segments –

i. Low

ii. Mid

iii. High

- This helps in understanding market segmentation.

4.2.3. KM Category – Based on kilometers driven –

i. Low Usage

ii. Medium Usage

iii. High Usage

- This improves analysis of vehicle condition and pricing behavior.

5. Key Performance Indicators (KPIs) –

- The following KPIs were identified to evaluate the business performance of the used car market –

i. Total Cars Listed – Measures dataset size and market volume

ii. Average Selling Price – Indicates overall market pricing

iii. Average Car Age – Shows vehicle age trends

iv. Most Preferred Fuel Type – Customer fuel preference

v. Percentage of Non-Accidental Cars – Vehicle condition indicator

vi. Total Revenue – Shows total revenue of the business

- These KPIs provide a quick summary of market behavior.

6. Exploratory Data Analysis (EDA) –

- Exploratory Data Analysis was performed using **Pivot Tables** and **Excel charts** to understand patterns and relationships in the data.

6.1 Price Trend Analysis –

i. Line charts were used to analyze **average price trends by year**
ii. Reveals how vehicle prices change over time

6.2 Brand Performance Analysis –

i. Column charts show **Top 10 brands by number of cars listed**
ii. Helps identify dominant brands in the resale market

6.3 Fuel Type Analysis –

i. Donut charts represent **fuel type distribution**
ii. Highlights customer fuel preferences

6.4 Car Type and Pricing –

i. Bar charts show **average price by car type**
ii. SUVs and Sedans generally command higher prices

6.5 Ownership & Accident Impact –

i. Donut and column charts analyze ownership type and accident history
ii. Non-accidental and first-owner cars dominate the market

8. Interactive Dashboard Design –

8.1 Dashboard Features –

i. KPI cards for summary metrics
ii. Multiple interactive charts
iii. Slicers for dynamic filtering

9. Tools and Technologies Used –

i. Kaggle Dataset
ii. Microsoft Excel
iii. Pivot Tables
iv. Charts
v. Slicers
vi. Formulas

10. Business Insights and Findings –

i. Maruti Suzuki and Hundai dominate the used car market

ii. Manual transmission vehicles are more common

iii. Petrol cars have the highest resale presence

iv. Car prices decrease as car age and kilometers increase

v. Non-accidental cars have significantly higher resale value

vi. Mid-range priced cars show the highest demand

11. Limitations of the Study –

i. Dataset is limited to available Kaggle records

ii. No real-time market pricing

iii. External factors like demand seasonality not included

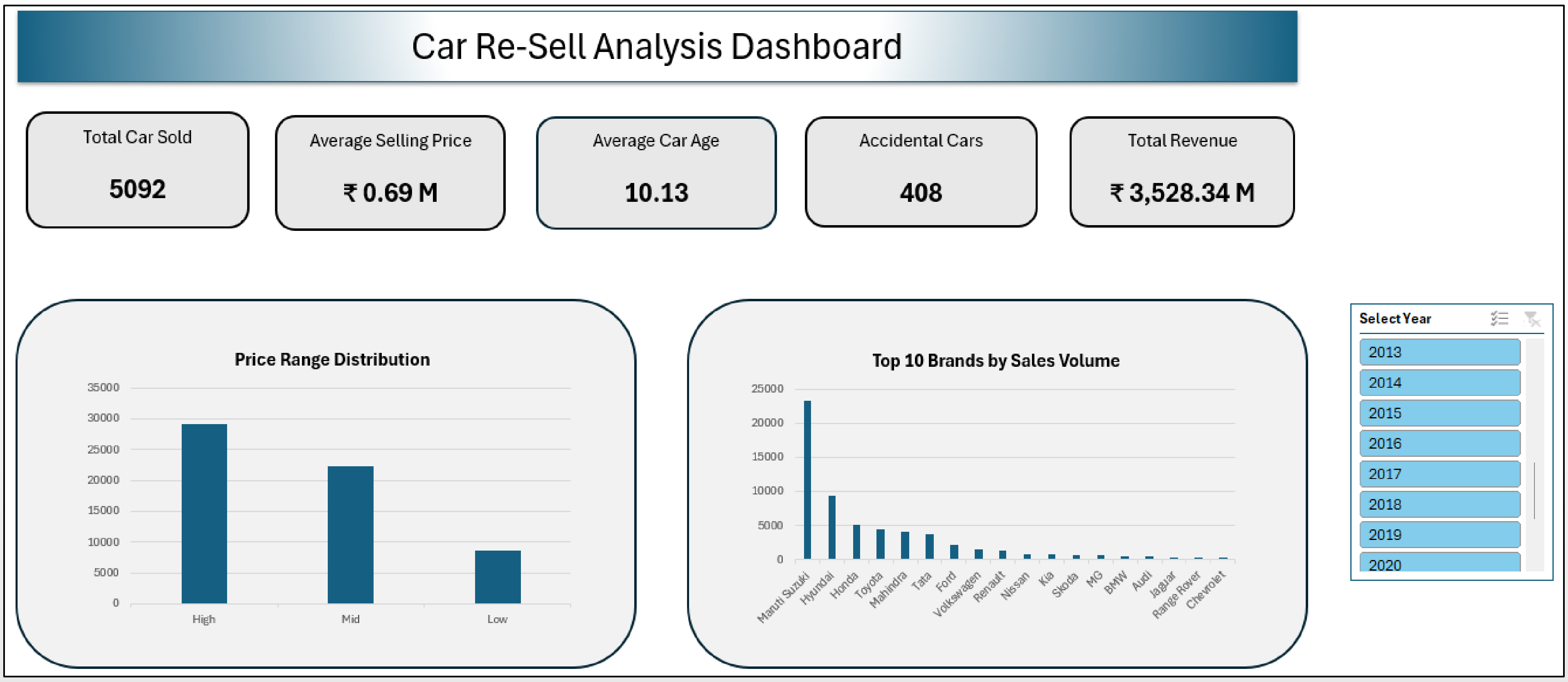
12. Conclusion –

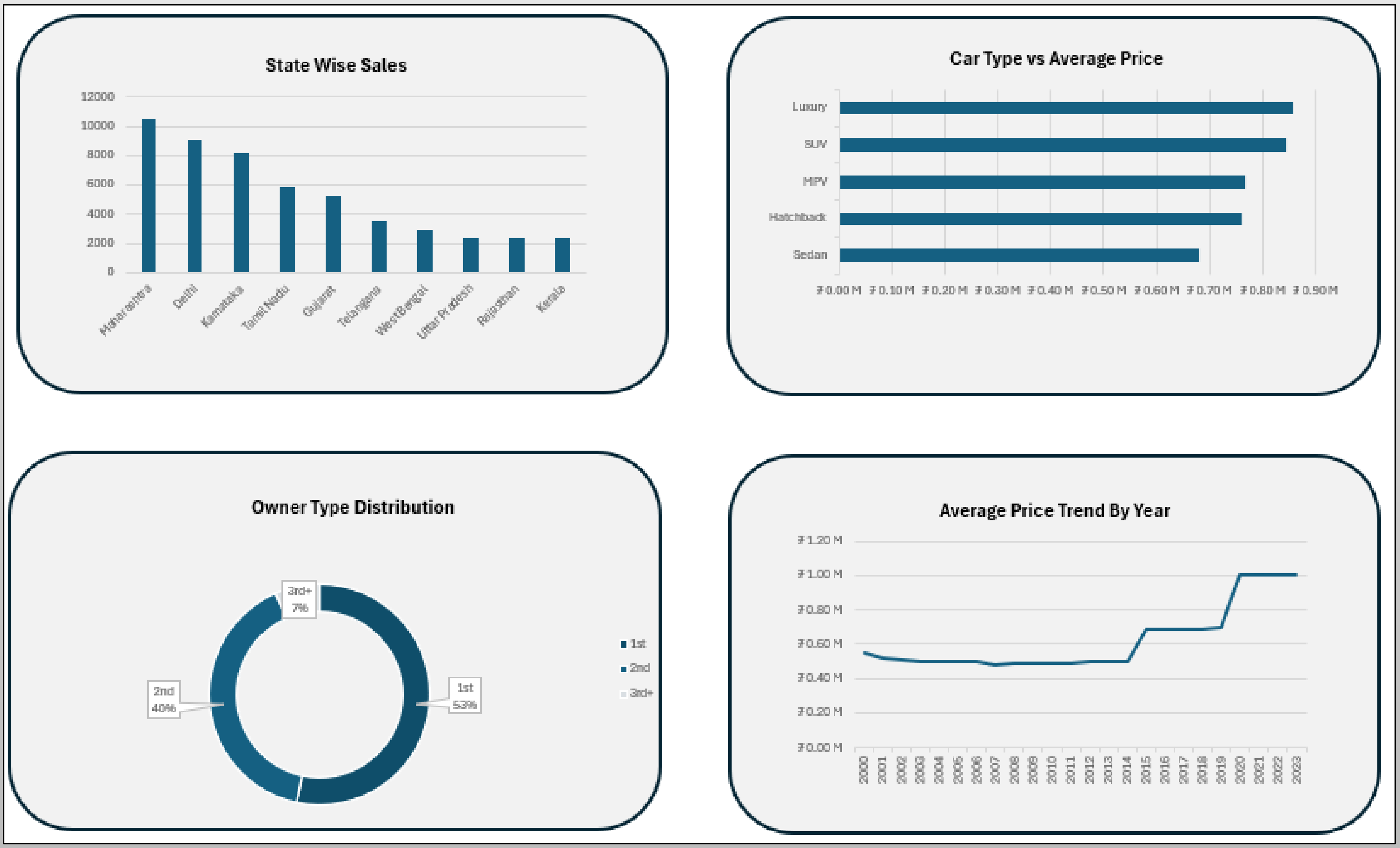
- This project successfully demonstrates how Microsoft Excel can be used to analyze a large real-world dataset and extract meaningful insights.

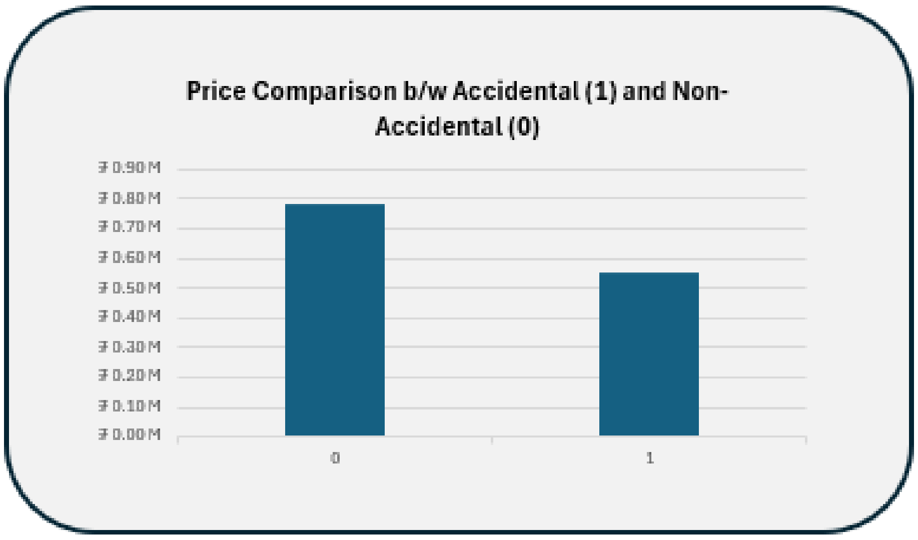
- Through data cleaning, KPI identification, pivot tables, visualizations, and an interactive dashboard, the project provides a comprehensive understanding of the used car resale market.

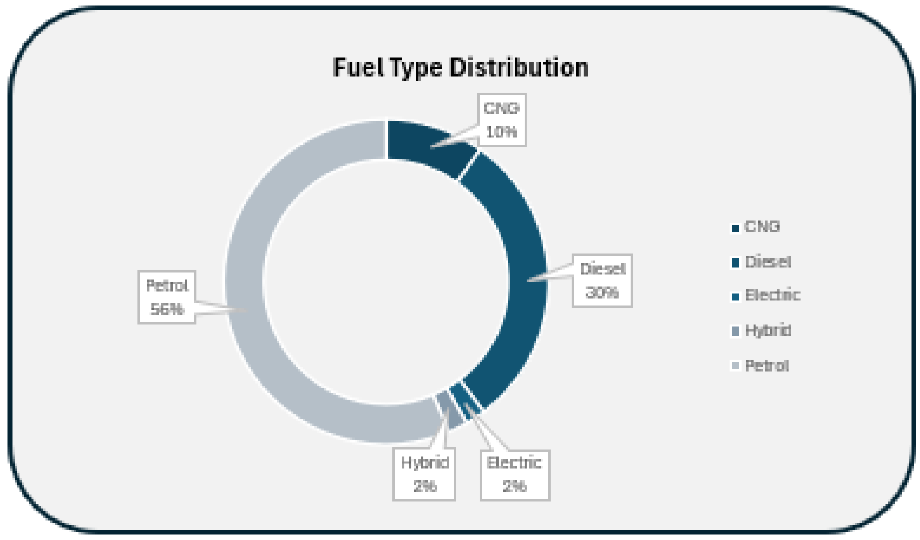
- The insights generated can help businesses optimize pricing, inventory planning, and customer targeting strategies.

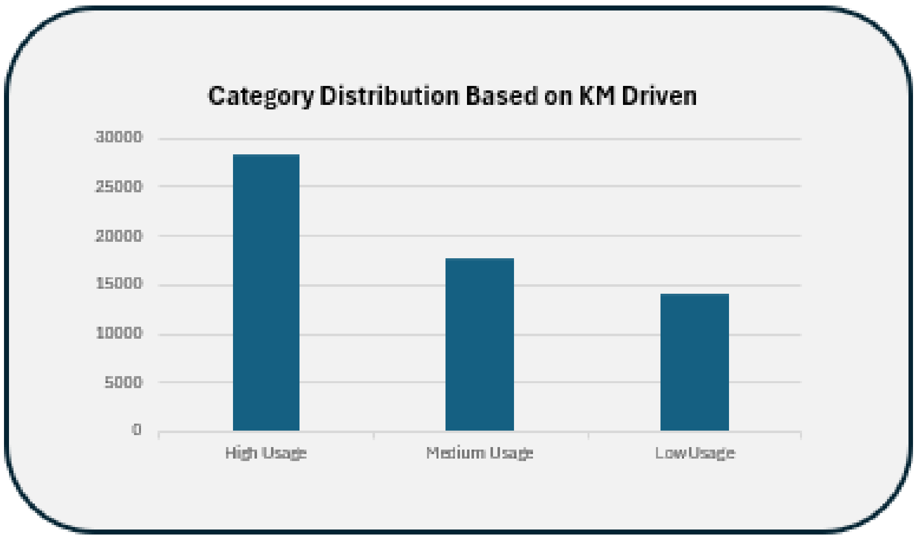
➢ Dashboard Pictures –

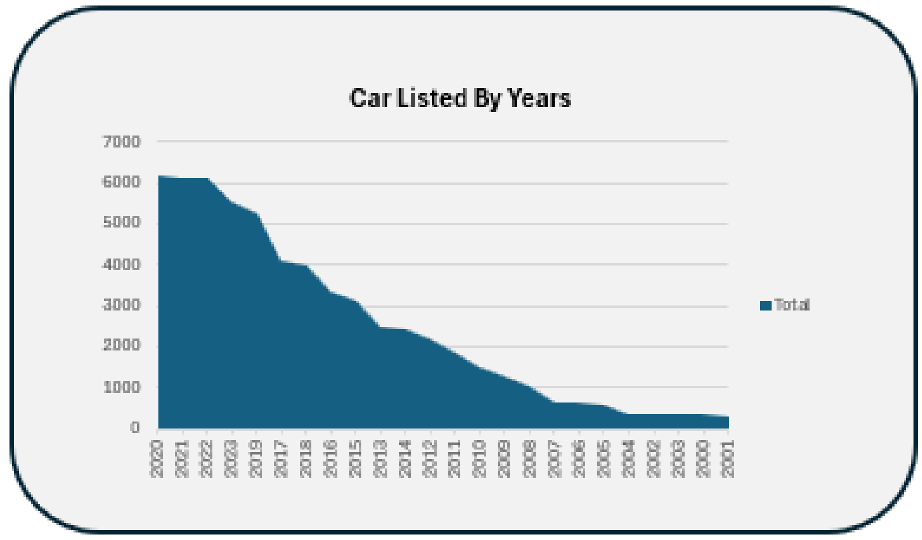












➢ KPI's –

Row Labels	Total Car Sold	Accidental Cars	Average Selling Price	Average Car Age	Total Revenue
Kia	902	54	₹ 1.02 M	4.31	₹ 922.67 M
MG	645	49	₹ 1.01 M	4.55	₹ 653.82 M
Tata	3714	280	₹ 0.90 M	5.60	₹ 3,350.68 M
Renault	1328	112	₹ 0.89 M	6.07	₹ 1,176.22 M
Range Rover	362	31	₹ 0.83 M	7.73	₹ 299.53 M
Toyota	4429	368	₹ 0.80 M	8.67	₹ 3,523.78 M
Jaguar	388	40	₹ 0.78 M	7.88	₹ 301.56 M
Skoda	671	52	₹ 0.78 M	8.86	₹ 521.21 M
Audi	437	32	₹ 0.78 M	8.68	₹ 339.36 M
Volkswagen	1565	116	₹ 0.77 M	8.08	₹ 1,197.70 M
Hyundai	9354	705	₹ 0.76 M	8.84	₹ 7,100.92 M
BMW	446	31	₹ 0.75 M	11.65	₹ 336.51 M
Maruti Suzuki	23252	1845	₹ 0.75 M	9.49	₹ 17,389.51 M
Nissan	922	78	₹ 0.73 M	9.07	₹ 670.89 M
Mahindra	4113	348	₹ 0.70 M	11.27	₹ 2,892.42 M
Honda	5092	408	₹ 0.69 M	10.13	₹ 3,528.34 M
Ford	2261	170	₹ 0.68 M	10.33	₹ 1,532.36 M
Chevrolet	326	20	₹ 0.55 M	13.28	₹ 178.10 M
Grand Total	**60207**	**4739**	**₹ 0.76 M**	**9.05**	**₹ 45,915.58 M**

➢ Power BI Dashboard

1. Introduction –

i. Cricket generates a massive volume of performance data across batting, bowling, fielding, and match conditions. With the growing popularity of fantasy sports platforms, there is a strong need to analyze this data in a structured and interactive manner to support better decision-making.

ii. This project focuses on building a **comprehensive, interactive Power BI dashboard** that analyzes tournament-level cricket data and translates it into actionable insights related to **player performance, team strength, match impact, and fantasy outcomes**.

iii. The dashboard is designed for both **analytical users** and **non-technical stakeholders**, providing high-level summaries as well as deep-dive analysis through interactivity.

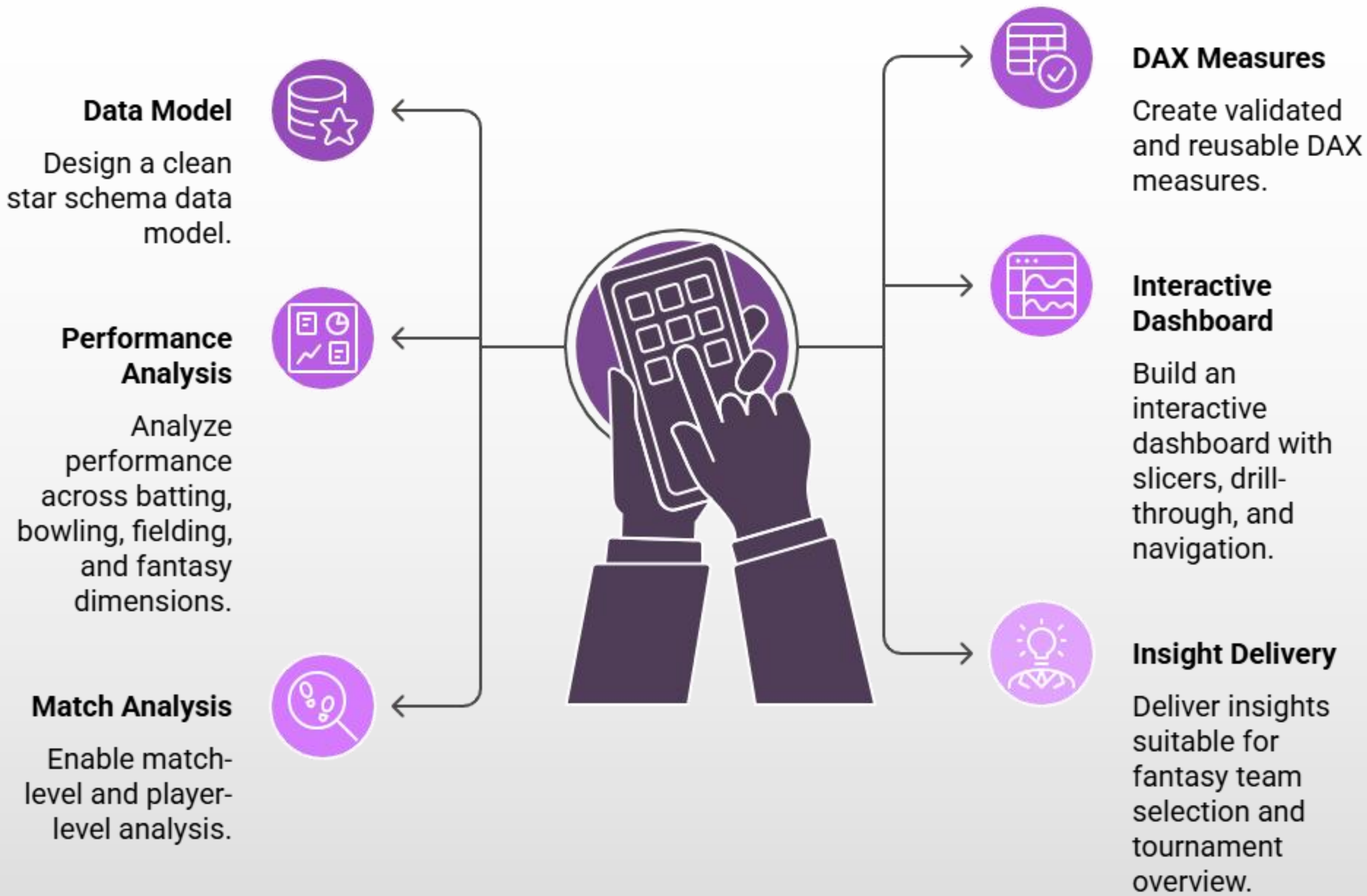
2. Business Problem Statement –

- Raw cricket data stored in tables is difficult to interpret due to: -

i. Lack of structure across batting, bowling, fielding, and fantasy datasets
ii. No centralized view of tournament performance
iii. Inability to compare players, teams, and matches interactively
iv. Difficulty in identifying fantasy-relevant players and consistent performers

- The objective of this project is to **transform raw cricket datasets into a unified analytical model** and present insights through a **single, interactive dashboard experience**.

3. Project Objectives –



4. Data Sources & Description –

- The project uses multiple structured datasets representing different aspects of the game –

i. **Batting Data** – Runs, balls, strike rate, boundaries

ii. **Bowling Data** – Wickets, economy, dot balls, overs

iii. **Fielding Data** – Catches, run-outs, fielding contributions

iv. **Fantasy Data** – Fantasy points, dream team selection

v. **Match Data** – Match ID, season, venue, match name

vi. Team Data – Team names

vii. Player Data – Player names

- Each dataset serves a distinct analytical purpose and is combined using relational modeling.

5. Data Cleening & Transformation (Power Query)

- Before analysis, the data was cleaned and prepared using **Power Query Editor**.

- Key transformation steps included –

i. Removing unnecessary columns

ii. Renaming fields for clarity and consistency

iii. Removing duplicate player, team, and venue values

iv. Creating dimension tables (Player, Match, Team, Venue)

v. Ensuring data types were correctly assigned

vi. Validating numerical fields such as balls, runs, and overs

- This ensured the dataset was **clean, standardized, and analysis-ready**.

6. Data Modeling (Star Schema Design) –

- A **star schema** was implemented to optimize performance and analytical clarity.

6.1. Dimension Tables –

i. Dim_Player
ii. Dim_Team
iii. Dim_Match
iv. Dim_Venue

6.2. Fact Tables –

i. Fact_Batting
ii. Fact_Bowling
iii. Fact_Fielding
iv. Fact_Fantasy

- Each dimension table connects to one or more fact tables using **one-to-many relationships** with **single-direction filtering**, ensuring accurate aggregations and avoiding ambiguity.

- This model supports –

i. Efficient filtering

ii. Accurate measure calculations

iii. Scalability for future data additions

7. Calculated Measures & DAX Logic –

- Custom DAX measures were created to derive meaningful metrics from raw data.

- These measures include –

i. Total Runs
ii. Total Wickets
iii. Average Strike Rate
iv. Average Economy Rate
v. Dot Ball Percentage
vi. Matches Played
vii. Total Fantasy Points
viii. Fantasy Points per Match
ix. Dream Team Percentage
x. Batting and Bowling Averages

- Each measure was validated to ensure correct aggregation behavior under slicer and filter contexts.

8. Dashboard Design Strategy –

- Instead of multiple disconnected dashboards, the project uses a **single, unified dashboard experience** built with –

i. Consistent layout
ii. Synced slicers
iii. Navigation buttons
iv. Role-based analysis pages

- This approach improves usability and mimics real-world BI applications.

9. Dashboard Pages & Analysis –

9.1 Home Page – Tournament Overview

- The home page acts as an **executive summary**, presenting –

i. Key KPIs (Runs, Wickets, Matches, Fantasy Points)
ii. Top fantasy performers
iii. Team-wise run distribution
iv. Fantasy trend across matches
v. Dream team distribution

- Navigation buttons allow seamless movement to detailed analysis pages.

9.2 Batting Analysis Page –

- This page focuses on batting performance insights such as –

i. Top run scorers
ii. Batting contribution by teams
iii. Match-wise batting trends
iv. Strike rate efficiency
v. Boundary analysis

- It helps identify **consistent and impactful batters**.

9.3 Bowling Analysis Page –

- The bowling analysis page highlights –

i. Top wicket-taking bowlers
ii. Economy efficiency
iii. Dot ball pressure
iv. Team bowling strength
v. Match-wise wicket trends

- This page is essential for understanding **bowling dominance and control**.

9.4 Dream Team Analysis Page –

- This page connects on-field performance with fantasy outcomes –

i. Top fantasy point earners
ii. Dream team appearances
iii. Fantasy contribution by team
iv. Match-wise fantasy trends
v. Dream vs non-dream team distribution

- It enables **fantasy-oriented decision making**.

10. Interactivity & Advanced Features –

- The dashboard includes –

i. Global slicers (Season, Team, Venue, Player)
ii. Synced slicers across pages
iii. Drill-through to match-level insights
iv. Page navigation using buttons
v. Context-aware filtering
vi. Responsive visual interactions

- These features enhance user experience and analytical depth.

11. Insights & Observations –

- Using the dashboard, users can –

i. Identify consistent performers across the tournament
ii. Compare team strengths in batting and bowling
iii. Analyze venue behavior
iv. Track performance trends over time
v. Make informed fantasy team selections

- The dashboard transforms raw data into **actionable insights**.

12. Tools & Technologies Used –

i. Power BI Desktop
ii. Power Query
iii. DAX (Data Analysis Expressions)
iv. Star Schema Modeling
v. Data Visualization Best Practices

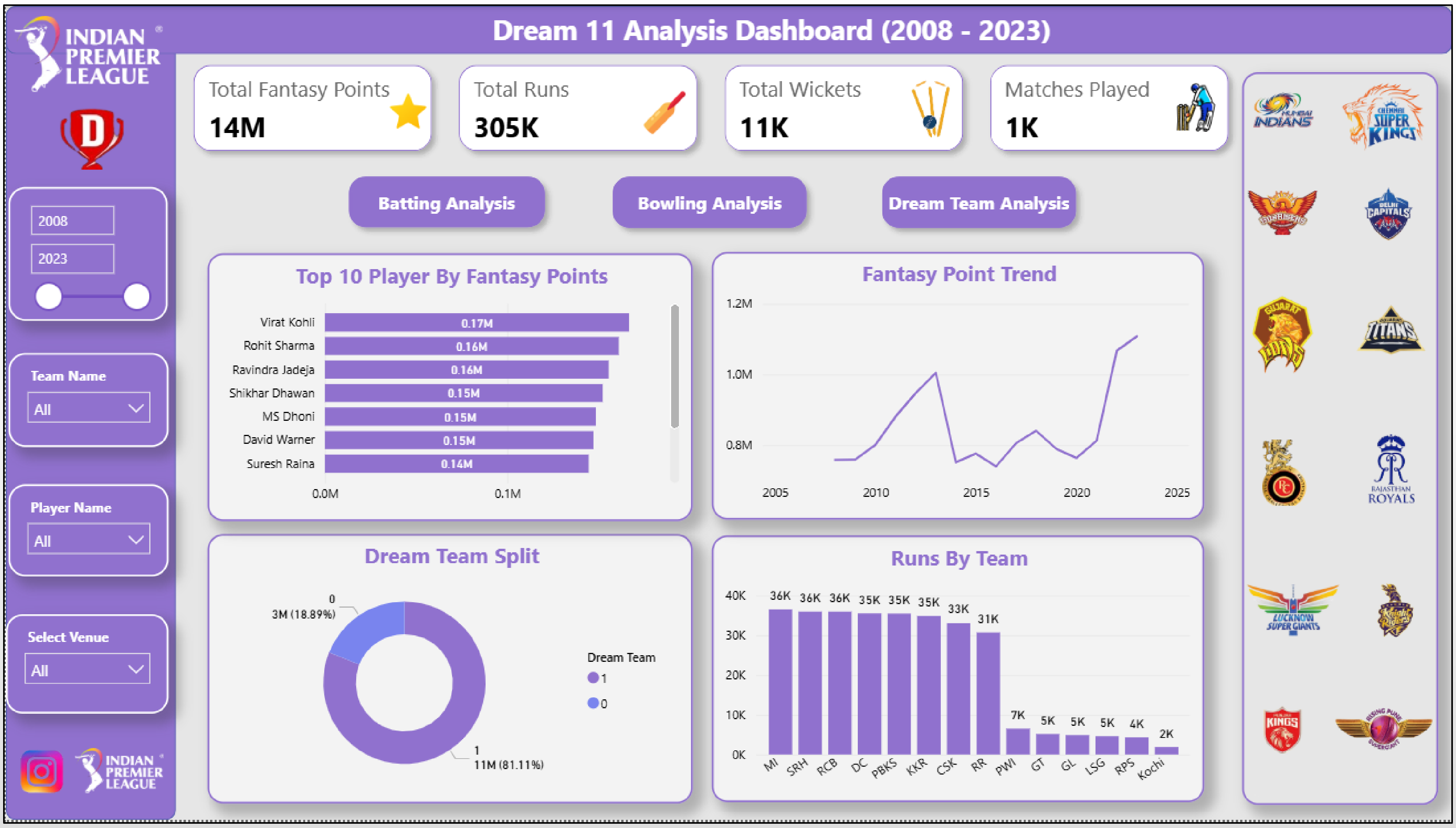
13. Conclusion –

- This project demonstrates an end-to-end data analytics workflow, from raw data ingestion to advanced visualization and insight generation.

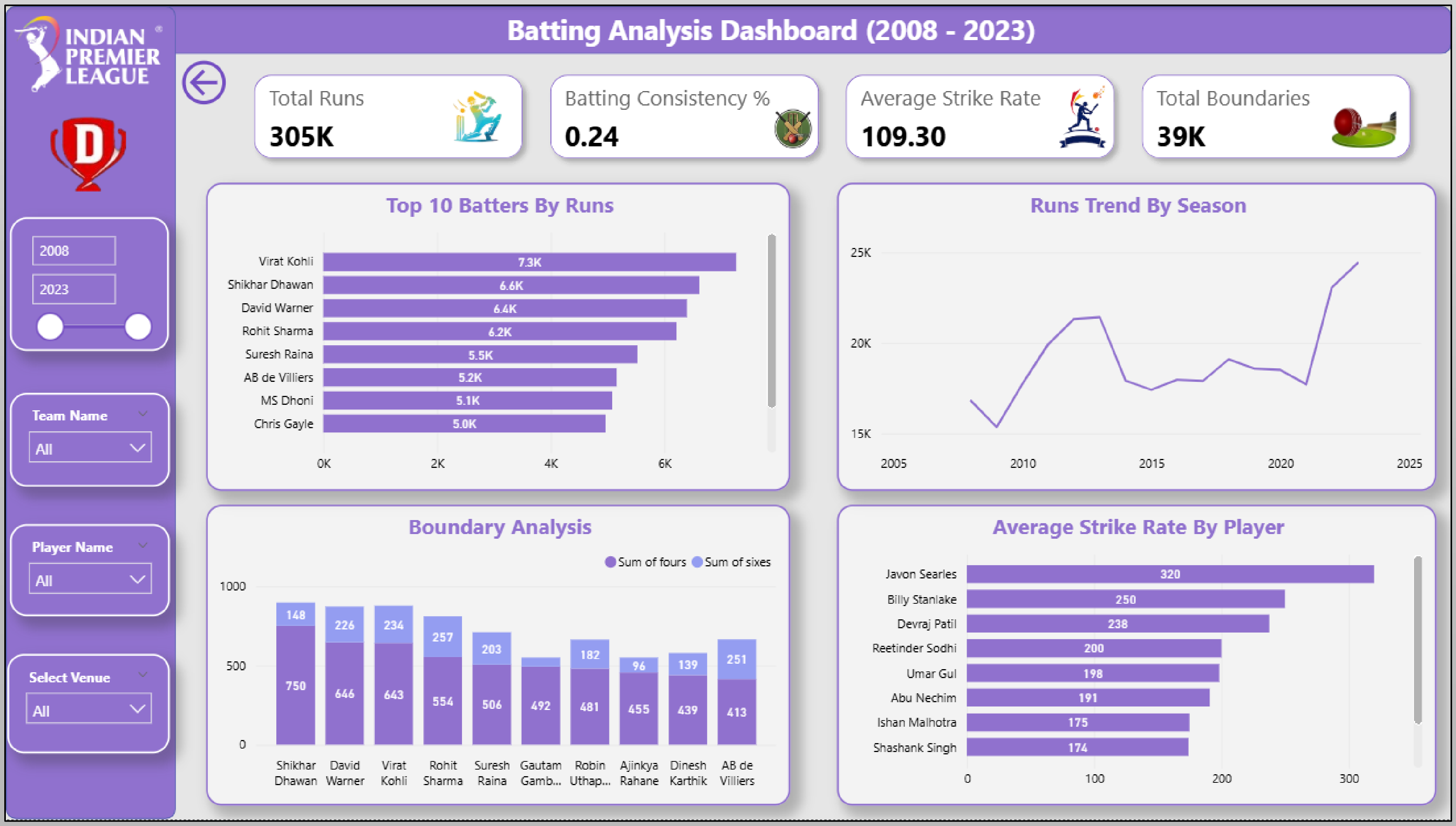
- By combining proper data modeling, validated DAX measures, and an interactive dashboard design, the project delivers a **scalable and professional analytics solution** suitable for real-world sports analytics and fantasy platforms.

➢ **Dashboard Pictures –**

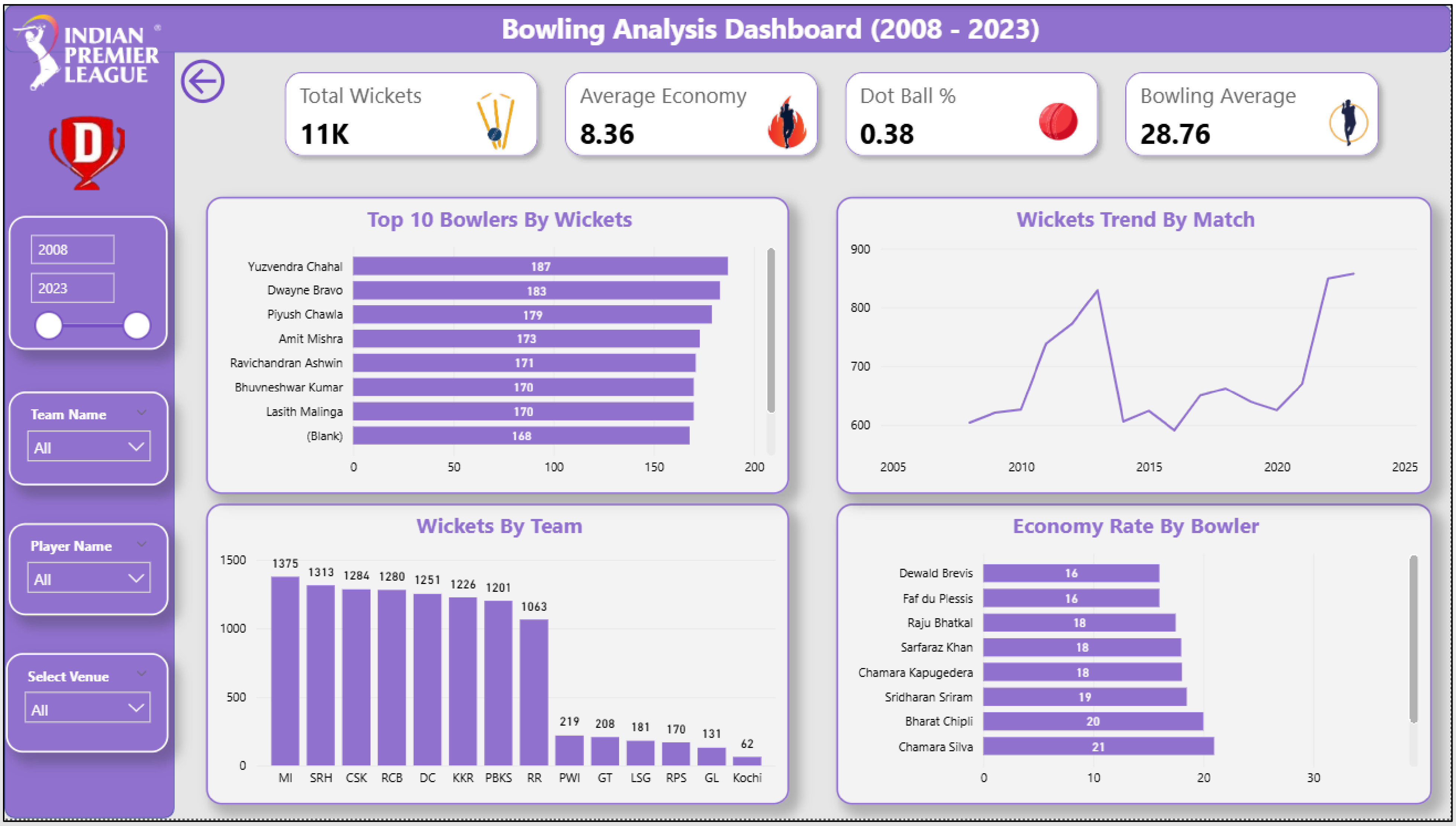
1. Home Page Dashboard –



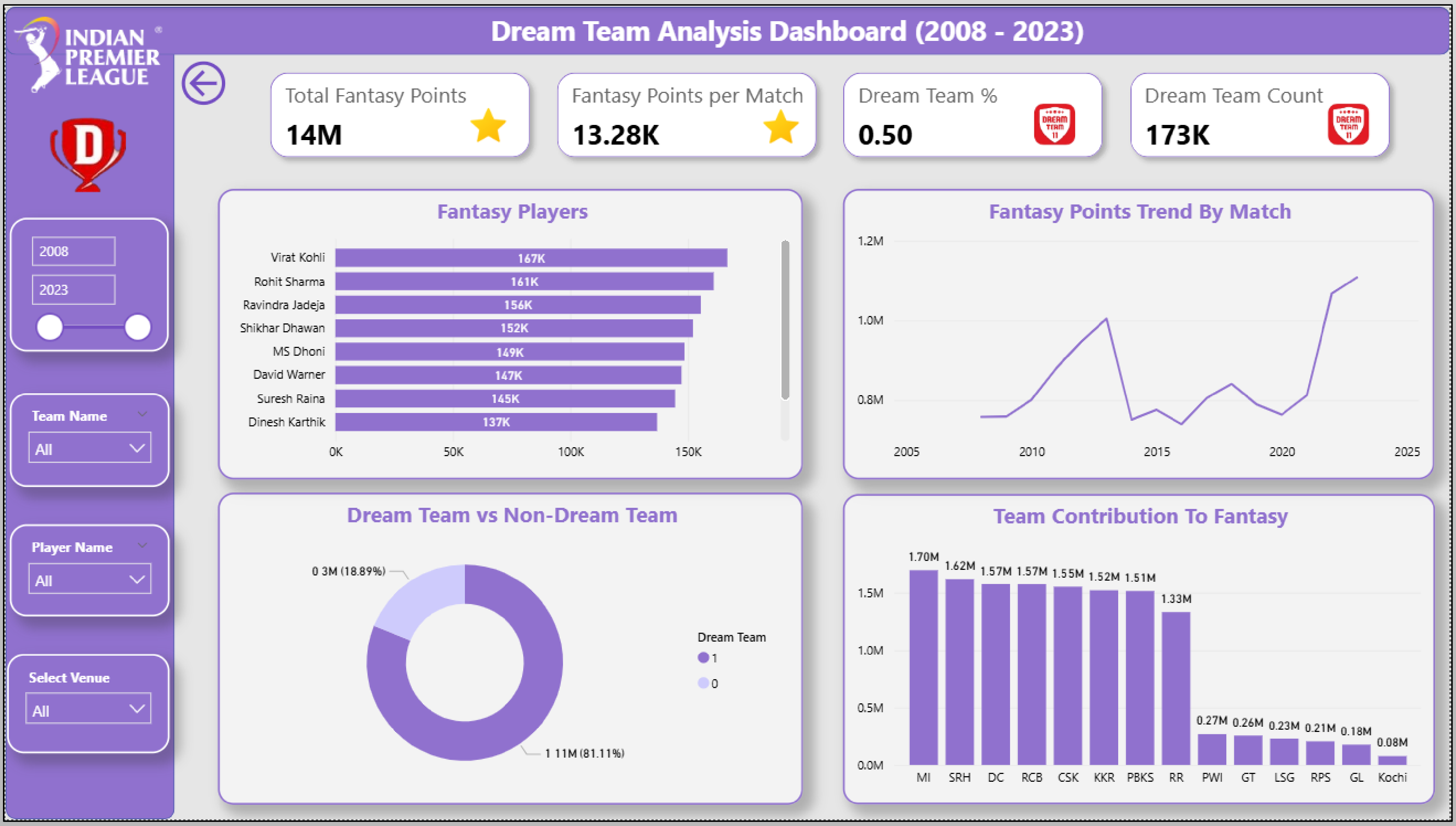
2. Batting Analysis Dashboard –



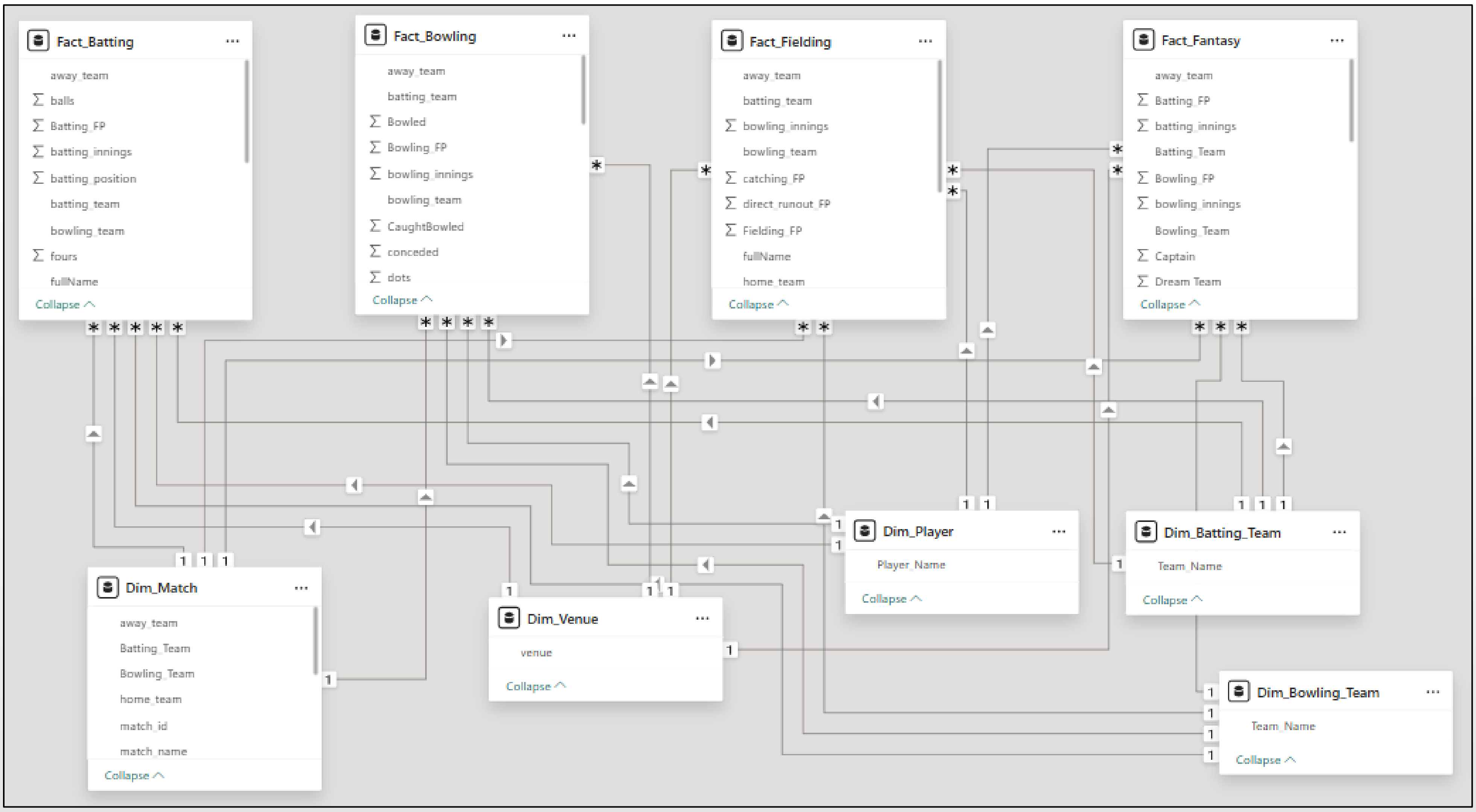
3. Bowling Analysis Dashboard –



4. Dream Team Analysis Dashboard –



5. Model View –



Fact_Batting
away_team
balls
Batting_FP
batting_innings
batting_position
batting_team
bowling_team
fours
fullName
Collapse
Fact_Bowling
away_team
batting_team
Bowled
Bowling_FP
bowling_innings
bowling_team
CaughtBowled
conceded
dots
Collapse
Fact_Fielding
away_team
batting_team
bowling_innings
bowling_team
catching_FP
direct_runout_FP
Fielding_FP
fullName
home_team
Collapse
Fact_Fantasy
away_team
Batting_FP
batting_innings
Batting_Team
Bowling_FP
bowling_innings
Bowling_Team
Captain
Dream Team
Collapse
Dim_Player
Player_Name
Collapse
Dim_Batting_Team
Team_Name
Collapse
Dim_Match
away_team
Batting_Team
Bowling_Team
home_team
match_id
match_name
Collapse
Dim_Venue
venue
Collapse
Dim_Bowling_Team
Team_Name
Collapse

14. Comparative Analysis of Data Visualization Tools –

14.1 Overview of the Tools –

i. Microsoft Excel is a widely adopted spreadsheet-based analytics tool used for data cleaning, exploratory analysis, KPI calculation, and basic dashboard creation. It is highly accessible and suitable for small to medium-sized datasets and quick analysis.

ii. Power BI / Tableau are modern Business Intelligence (BI) platforms designed for large-scale data modeling, advanced analytics, interactive dashboards, and enterprise-level reporting. Power BI, in particular, integrates data modeling, DAX calculations, and visualization in a unified environment.

14.2 Data Handling & Scalability –

i. Excel performs efficiently for small to moderately sized datasets but faces performance limitations as data volume increases. Complex calculations, large pivot tables, and heavy visualizations may slow down processing.

ii. Power BI and Tableau are built to handle large datasets efficiently. The use of **columnar storage, in-memory engines, and optimized data models (star schema)** allows these tools to process millions of rows smoothly.

✓ **Comparison Insight –** Excel is ideal for static or moderate datasets, whereas Power BI/Tableau are more suitable for scalable, enterprise-level analytics.

14.3 Data Modeling Capability –

i. Excel relies primarily on flat tables and relationships through lookup formulas or Power Pivot (optional). While functional, it lacks the robustness of a full semantic model.

ii. Power BI/Tableau support **relational data models**, including fact and dimension tables, defined relationships, and filter directions. Power BI's star schema implementation significantly improves performance and analytical accuracy.

✓ **Comparison Insight –** Power BI provides a more structured and professional data modeling environment compared to Excel.

14.4 Analytical & Calculation Features –

i. Excel uses formulas, pivot table calculations, and basic measures. While powerful, complex calculations can become difficult to maintain and audit.

ii. Power BI introduces **DAX (Data Analysis Expressions)**, enabling reusable, context-aware measures that dynamically respond to filters and slicers. Tableau provides calculated fields with similar flexibility.

✓ **Comparison Insight –** Power BI/Tableau offer advanced, reusable, and scalable calculation frameworks, whereas Excel is better suited for simpler KPI-based analysis.

14.5 Interactivity & User Experience –

- Excel dashboards provide limited interactivity through slicers and pivot charts. User experience is mostly static and sheet-based.

- Power BI/Tableau dashboards offer –

i. Cross-filtering visuals
ii. Drill-through analysis
iii. Page navigation
iv. Synced slicers
v. Context-aware interactions

- This enables deeper exploration without altering the underlying data.

✓ **Comparison Insight –** Power BI/Tableau deliver a significantly richer and more intuitive interactive experience than Excel.

14.6 Use Case Alignment (Based on Projects) –

Aspect	Used Car Sales Analysis (Excel)	Cricket Fantasy Analytics (Power BI)
Dataset Size	Medium	Large & multi-table
Data Model	Flat / Pivot-based	Star schema
Analytics Depth	Descriptive & KPI-based	Advanced performance & fantasy analytics
Interactivity	Limited	High
Target Audience	Analysts & business users	Analysts, stakeholders, fantasy users

14.7 Strengths & Limitations Summary –

1. **Microsoft Excel – Strengths –**

i. Easy to learn and widely available

ii. Ideal for quick analysis and EDA

iii. Excellent for prototyping and small dashboards

2. **Microsoft Excel – Limitations –**

i. Limited scalability

ii. Restricted interactivity

iii. Not ideal for complex data models

3. **Power BI / Tableau – Strengths –**

i. Advanced data modelling and calculations

ii. High interactivity and storytelling

iii. Scalable and enterprise-ready

4. **Power BI / Tableau – Limitations –**

i. Steeper learning curve

ii. Licensing considerations

15. Conclusion –

- Both Microsoft Excel and Power BI/Tableau play important roles in the data analytics ecosystem. Excel serves as a strong foundation for data cleaning, exploratory analysis, and basic dashboards, while Power BI/Tableau enable advanced analytics, scalable data modeling, and interactive business intelligence solutions.

- The two projects collectively demonstrate the ability to **choose the right tool for the right problem**, showcasing versatility, technical understanding, and real-world analytics readiness.